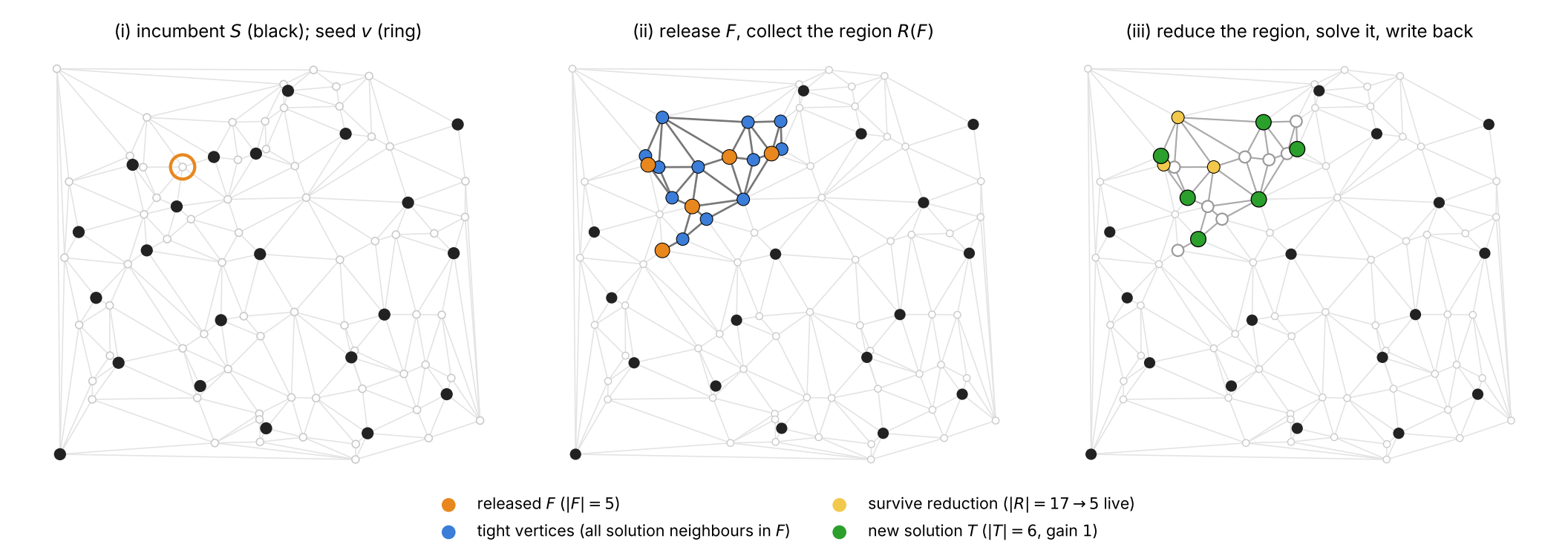
(i) incumbent S (black); seed v (ring)
(ii) release F, collect the region $R(F)$
(iii) reduce the region, solve it, write back
released F ($|F| = 5$)
tight vertices (all solution neighbours in F)
survive reduction ($|R| = 17 \to 5$ live)
new solution T ($|T| = 6$, gain 1)